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Time Series Analysis and Forecasting of Exchange Rates in Lebanon

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Abstract

This project examines the analysis and forecasting of the Lebanese Pound (LBP) exchange rate against the US Dollar from 2019 to 2024. Both ARIMA and ETS (Exponential Smoothing) models were applied for univariate time series analysis, with ETS showing better accuracy based on Mean Absolute Error (MAE), Root Mean Square Error (RMSE).

To further improve forecasting, ARIMAX models were implemented by incorporating external variables such as interest rates, foreign currency reserves, and the Consumer Price Index (CPI). This multivariate approach demonstrated enhanced predictive performance over ARIMA by effectively capturing the influence of external factors. Key steps included transforming non-stationary data, identifying ARIMA orders through ACF and PACF plots, checking for white noise, and validating models using metrics such as AIC and BIC.

These findings underscore the importance of combining univariate and multivariate approaches, as well as external economic indicators, to effectively forecast exchange rate dynamics and support decision-making in volatile financial contexts.

Keywords

Time Series Analysis, ARIMA, ARIMAX, ETS, Holt's method, Holt's winter, KPSS, Exchange Rates, Lebanon, Forecasting, Economic Indicators

Introduction

Since 2019, Lebanon has been grappling with a profound economic crisis, leading to the Lebanese Pound (LBP) losing over 98% of its value against the US Dollar (L'Orient Today, 2024). This drastic devaluation has resulted in multiple exchange rates and significant market distortions, severely impacting economic stability and public confidence (NOW Lebanon, 2024). In February 2023, the Central Bank of Lebanon devalued the official exchange rate by 90%, yet the gap with the black-market rate persisted (Independent, 2023). Given this volatility, accurate exchange rate forecasting has become essential for policymakers, investors, and businesses to navigate the uncertain financial landscape.

Forecasting exchange rates provides insights into future currency behavior, helping stakeholders manage risks and make informed decisions. Time series models, such as ARIMA (AutoRegressive Integrated Moving Average), ETS (Exponential Smoothing), and ARIMAX (ARIMA with external variables), are widely used in financial modeling due to their ability to capture patterns and trends in historical data (*Hyndman & Athanasopoulos, 2018*).

ARIMA is particularly effective for univariate time series analysis, while ETS excels at modeling seasonality and trend components (Box et al., 2015).

The primary objective of this study is to analyze and forecast the LBP exchange rate against the US Dollar using these models. By comparing the performance of univariate models (ARIMA and ETS) with the multivariate ARIMAX approach, the study aims to evaluate the significance of external economic indicators in improving forecast accuracy. Additionally, the research highlights the value of combining statistical methods with external variables to provide robust and actionable insights for navigating Lebanon's exchange rate volatility.

Data Description

The dataset used for this study was created by combining four distinct datasets to provide comprehensive coverage of the Lebanese Pound (LBP) exchange rate and its influencing factors. The datasets span the time period from 2019 to 2024 and were sourced from reliable institutions.

Exchange Rates

The primary dataset includes the monthly exchange rates (Open, Close, Low, and High) and was obtained from the World Bank's dataset titled *Lebanon - Monthly Currency Exchange Rate Estimates by Market* (World Bank, 2024). Among these, the **Close rate** was selected as the representative exchange rate for each month to maintain consistency and focus.

Predictor Variables

Three additional datasets were sourced from the Banque du Liban:

- **Foreign Currency Reserves**: Monthly reserves in USD, representing Lebanon's capacity to stabilize its currency. Represented in billions of dollars.
- **Interest Rates**: USD deposit average interest rates in Lebanese banks.
- **Consumer Price Index (CPI)**: A measure of inflation reflecting changes in the cost of living.

Normalization

To simplify the representation of the exchange rates, the values were divided by 1,000 since the original data was in thousands. For instance, an exchange rate of 1,500 LBP/USD was recorded as 1.5 in the dataset.

Preprocessing and Alignment:

All datasets were aligned to match the monthly time series from 2019 to 2024. Missing data points, if any, were addressed through interpolation to ensure a consistent

dataset. Additionally, the external variables were scaled and transformed where necessary to match the exchange rate dataset's scale and facilitate seamless integration into the analysis.

Choice of Variables

The choice of variables for this study—interest rates, foreign currency reserves, and the Consumer Price Index (CPI)—is based on their widely recognized influence on exchange rates. Higher interest rates attract foreign investment, leading to currency appreciation, while lower rates cause depreciation (Investopedia, 2023). Foreign currency reserves play a crucial role in stabilizing exchange rates by enabling market interventions during volatility (BIS, 2021). Lastly, the CPI reflects inflation, which directly impacts purchasing power and currency value; higher inflation typically leads to depreciation (Investopedia, 2023). Including these variables enhances the robustness of the analysis and captures critical economic factors affecting the LBP/USD exchange rate.

Methodology:

Exploratory Data Analysis (EDA):

This section focuses on analyzing the characteristics of the exchange rate and external variables before applying any preprocessing.

Overview of Exchange Rate, Foreign Currency Reserves, and CIP Trends:

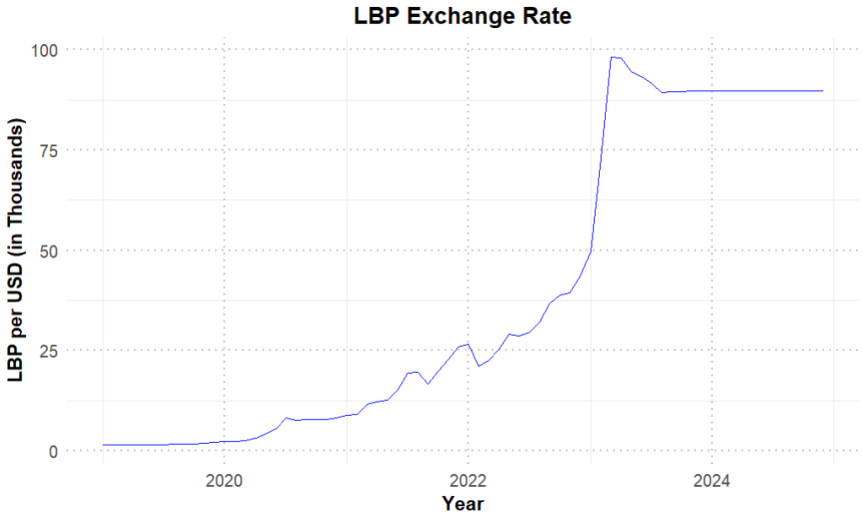


Figure 1: LBP Exchange Rate

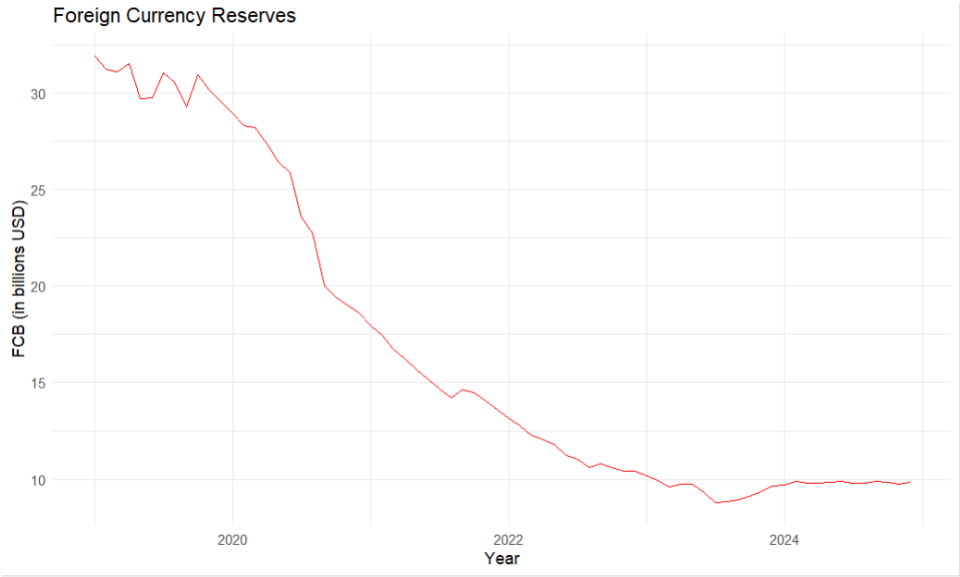


Figure 2: Foreign Currency Reserves in Billions

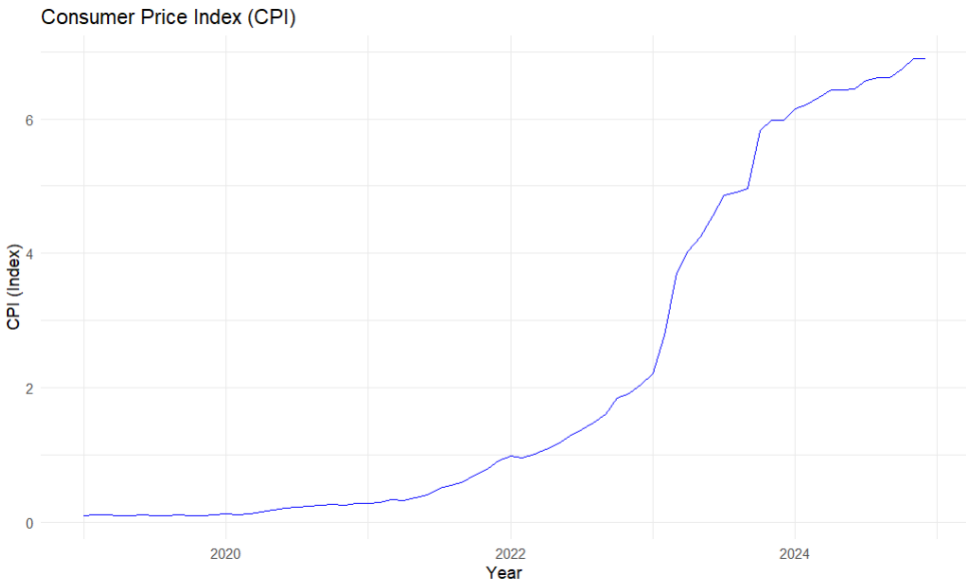


Figure 3: CIP Rates

The three graphs collectively illustrate interconnected economic trends in Lebanon. The sharp depreciation of the Lebanese Pound (LBP) against the US Dollar, as shown in the first graph, is accompanied by a significant decline in foreign currency reserves (FCB) and a steep rise in the Consumer Price Index (CPI). The depletion of FCB, particularly after 2020, likely contributed to a loss of confidence in the LBP, fueling its depreciation. Simultaneously, the rapid increase in CPI highlights growing inflation, further reducing the purchasing power of the currency. These trends suggest a strong interrelationship between the declining reserves, inflation, and exchange rate instability, emphasizing the importance of including

FCB and CPI as regressors in the ARIMAX model to better capture and forecast exchange rate dynamics.

Statistical Summary:

Table 1: Descriptive Statistics

Data	Minimum	1st Quartile	Mean	Standard Deviation	3rdQuartile	Maximum
Exchange Rate	1.515	6.964	38.553	37.3108389.7	89.7	98.265
CIP	0.09637	0.23104	2.21006	2.51479	4.62678	6.89631
Foreign Currency Reserves	8.759	9.823	16.785	8.165763	31.932	
Interest Rates	0.0200	0.06775	0.1850	2.166458	1.5275	6.6100

Preprocessing

Train-Test Split

Training Set:

- Contains data from January 2019 to December 2023.
- Includes a total of **60 observations** (5 years of monthly data).

Testing Set:

- Contains data from January 2024 onwards.
- Includes a total of **12 observations** (1 year of monthly data).

Correlation Analysis:

The correlations between the exchange rate, CPI, foreign currency reserves (FCB), and interest rates were computed using **Pearson's correlation** coefficient to evaluate the strength of their linear relationships. This step is essential for selecting the most relevant explanatory variables to include in the ARIMAX model. Variables with strong correlations to the dependent variable (exchange rate) are likely to contribute significantly to the model's predictive power, while those with weaker correlations may have minimal impact and can be excluded to simplify the model (Hyndman & Athanasopoulos, 2018).

Variable	Exchange Rate	CPI	FCB	Interest Rate
Exchange Rate	1.000	0.960	−0.786	−0.599
CPI	0.960	1.000	−0.700	−0.517
FCB	−0.786	−0.700	1.000	0.919
Interest Rate	−0.599	−0.517	0.919	1.000

Based on the correlation analysis, **CPI** and **FCB** were chosen as explanatory variables for the ARIMAX model due to their strong correlations with the exchange rate (r=0.960 and r=−0.786, respectively). The **interest rate** was excluded because its correlation with the exchange rate (r=−0.599r) is relatively weaker, indicating limited predictive value.

No significant autocorrelation was observed between the regressors (**CPI** and **FCB**). A moderate relationship that does not pose multicollinearity concerns.

Transformations:

To stabilize the variance and prepare the variables for analysis, the Box-Cox transformation was applied to the exchange rate, CPI, and foreign currency reserves. The Box-Cox method transforms non-linear data into a form that is closer to normal distribution, making it suitable for time series modeling (Box & Cox, 1964). The transformation is defined as:

$$y(\lambda) = \begin{cases} \frac{y^{\lambda}-1}{\lambda}, & \text{if } \lambda \neq 0 \\ \ln(y), & \text{if } \lambda = 0 \end{cases}$$

Here, λ is a parameter estimated to minimize the log-likelihood function of the transformed data. For the exchange rate, a λ value of 0.302 was obtained, Similar transformations were performed for the CPI and foreign currency reserves to ensure consistency in the analysis and improve model performance by addressing heteroscedasticity.

Stationary Tests:

Stationarity is a critical assumption for time series modeling, as non-stationary data can lead to biased parameter estimates and unreliable forecasts. To test for stationarity, the KPSS (Kwiatkowski-Phillips-Schmidt-Shin) unit root test was applied to the transformed variables, including the exchange rate, CPI, and foreign currency reserves.

The KPSS test evaluates the null hypothesis (H_0) that the series is stationary against the alternative hypothesis (H_1) of non-stationarity. The test statistics are calculated as:

$$\eta = T^{-2} \sum_{t=1}^{T} S_t^2 / \hat{\sigma}^2$$

where $\mathbf{S_t}$ is the cumulative residual, **T** is the sample size, and σ^2 is the long-run variance. If the test statistic exceeds the critical value, the null hypothesis of stationarity is rejected (Kwiatkowski et al., 1992).

For the transformed exchange rate (λ=0.302), the KPSS test yielded a test statistic of 1.5389, exceeding the 1% critical value of 0.739, indicating non-stationarity. To achieve stationarity, first-order differencing was applied, as suggested by the ndiffs() function. After differencing, the KPSS test was repeated, yielding a test statistic of 0.1152, below the 5% critical value of 0.463, confirming stationarity. Seasonal differencing was unnecessary, as nsdiffs() returned 0 for all variables.

By transforming and differentiating the data, all variables were rendered stationary, meeting the requirements for effective time series modeling.

ARIMA Modeling for Exchange Rate Forecasting

The ARIMA (AutoRegressive Integrated Moving Average) model was employed to forecast the Lebanese Pound (LBP) exchange rate against the US Dollar. ARIMA is a statistical time series modeling technique that combines three key components—autoregression (AR), differencing (I), and moving average (MA)—to capture temporal dependencies and trends in univariate data. The general form of the ARIMA model is:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q}$$

where Y_t is the differenced time series, ϕ are AR coefficients, θ are MA coefficients, and ϵ_t represents white noise.

Model Identification

To identify appropriate ARIMA parameters (*p*, *d*, *q*), the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were analyzed:

- **ACF Plot**: Indicated significant lag at 1, suggesting ***q***=1.
- **PACF Plot**: Indicated significant lag at 1, suggesting **p**=1.
- First-order differencing (***d***=1) was applied to address non-stationarity, as confirmed by the KPSS test.

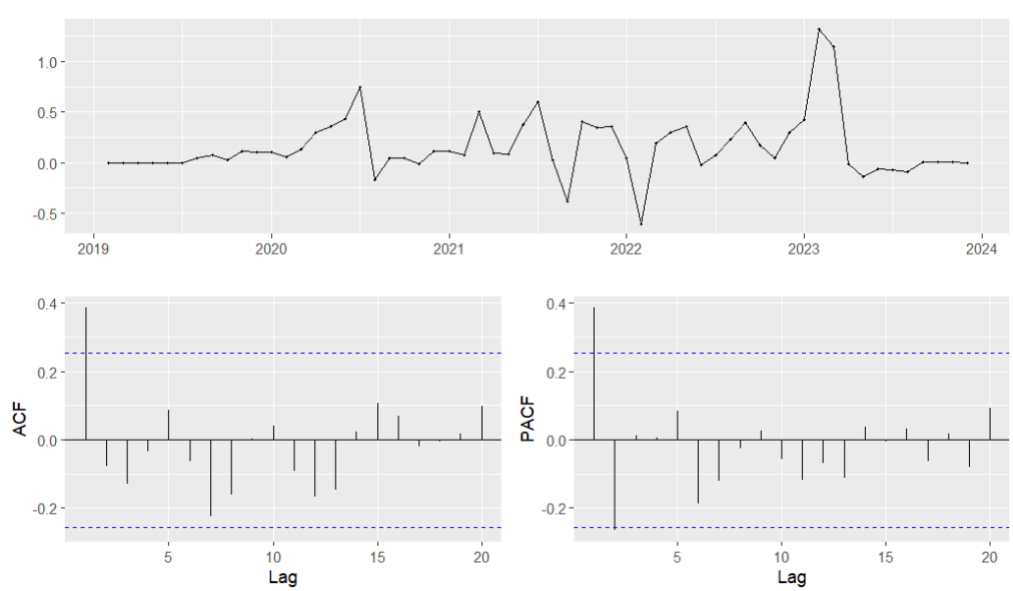


Figure 4: ACF and PACF for Exchange Rate

Model Estimation and Comparison

Five ARIMA models were tested, and their performance was evaluated using the corrected Akaike Information Criterion (AICc). The results are summarized in Table 2.

Table 2: AIC measures for the ARIMA models

Models	AICc
Arima (1,1,0)	24.735
Arima (1,1,1)	24.963
Arima (0,1,1)	23.648
ARIMA (1,1,1) + Drift	24.963
Arima (0,1,1) + Drift	**18.064**

The **ARIMA (0,1,1) with drift** model was selected as the best model due to its lowest AICc value.

Coefficients of ARIMA (0,1,1) with Drift:

$$Y_t = 0.153 + (1 - 0.503B)^{-1}\epsilon_t$$

Model Diagnostics

The adequacy of the selected model was assessed using residual diagnostics: Ljung-Box, ACF Plot of Residuals, and Histogram of Residuals.

The **Ljung-Box** test returned a p-value of 0.864, which is greater than the significance level of 0.05. This indicates that we fail to reject the null hypothesis, confirming the residuals are white noise and the model sufficiently captures the data structure (Ljung & Box, 1978). As shown in figure 3:

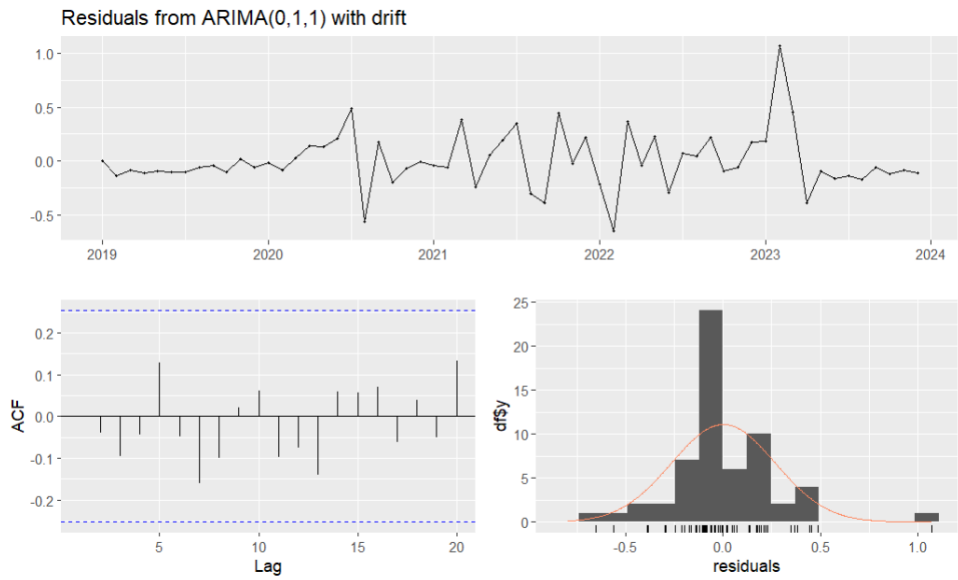


Figure 5: Analysis of Residuals

- **ACF Plot of Residuals:** No significant autocorrelation was observed.
- **Histogram of Residuals:** Residuals were approximately normally distributed, with zero mean.

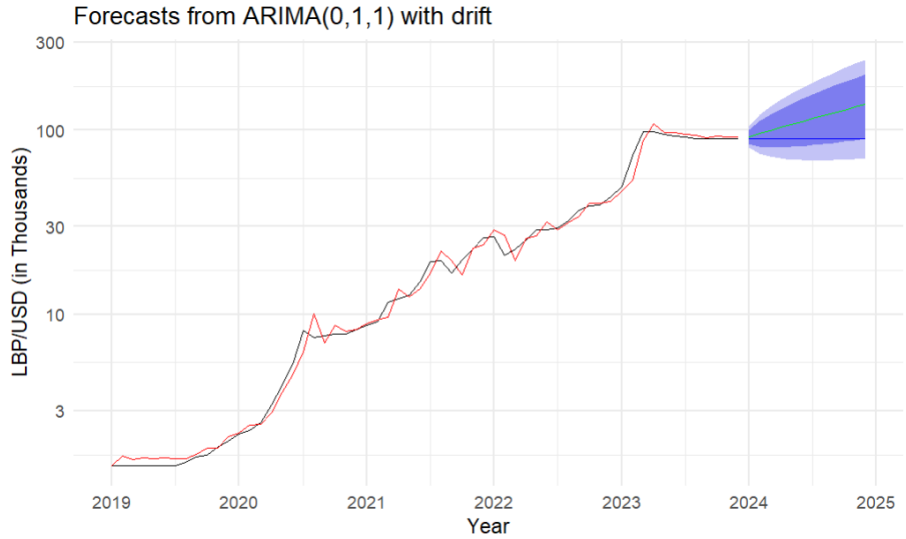


Figure 6: Forecast Graph of ARIMA (0,1,1) with drift

Accuracy Measures for ARIMA (0,1,1) with drift for the train and test sets:

Train Set:

- **Mean Absolute Error (MAE):** 1.98
- **Root Mean Squared Error (RMSE):** 3.66
- **Mean Absolute Percentage Error (MAPE):** 8.16%
- **Mean Error (ME):** 0.02
- **ACF1:** 0.25

Test Set:

- **MAE:** 24.17
- **RMSE:** 28.11
- **MAPE:** 26.95%
- **Mean Error (ME):** -24.17
- **ACF1:** 0.75

These measures indicate that while the model performs well on the training set, the test set errors are considerably larger, reflecting the challenges in forecasting the highly volatile exchange rate.

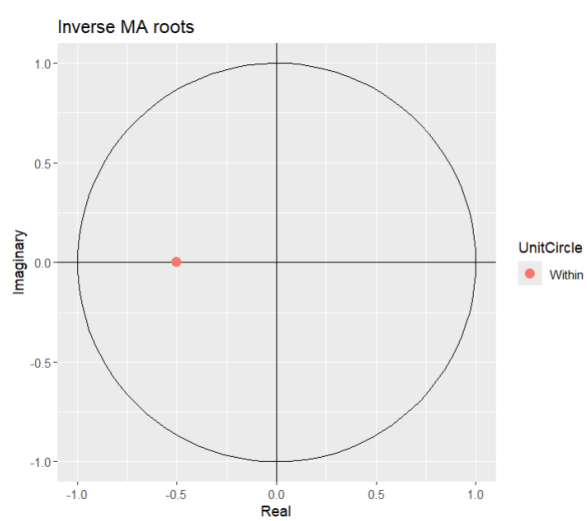


Figure 7: Inverse Characteristic

The red dot lies inside the unit circle. Hence, the fitted model is both stationary and invertible.

ETS Modeling for Exchange Rate Forecasting:

1. Simple Exponential Smoothing (SES): This model is suitable for data without a trend or seasonality. It uses a smoothing parameter (α) to assign exponentially decreasing weights to older observations.

2. Holt's Method: This method extends SES by incorporating a trend component, using two smoothing parameters (α for the level and β for the trend).

3. Holt-Winters Method: This method builds on Holt's by adding a seasonal component. It can be applied in two variants:

- **Additive:** Suitable for data where the seasonality remains constant over time.
- **Multiplicative:** Suitable for data where the seasonal effect increases or decreases proportionally to the level of the series.

Results of ETS Models:

Model	AICc	RMSE (Training)	RMSE (Testing)	MAPE (%) (Test)	MAPE (%) (Test)
Simple Exponential Smoothing (SES)	442.35	4.88	0.00	8.14	0.00
Holt's Method	441.93	4.68	15.30	12.73	15.07
Holt's (Damped)	431.76	4.45	19.43	39.00	20.20
Holt-Winters (Additive)	473.29	5.51	54.68	24.63	56.41

Table 3: Results for ETS Models

Among the ETS models tested, **Holt's Method** demonstrated the best balance between training accuracy and model simplicity, while SES offered the most stable short-term forecasts. However, all ETS models struggled to capture the extreme volatility of the

exchange rate, highlighting the need for more complex models in such cases. Future improvements could involve incorporating external variables or nonlinear modeling techniques.

Ljung-Box Test and ACF plots were used to analyze the residuals to validate the model/ All the methods indicated marginal white noise and the ACF plot showed a minimal autocorrelation, confirming the model captured the underlying structure adequately.

Forecasting

As shown from the figure below, the models were used to forecast the LBP/USD exchange rate for 12 months. The SES model provided stable forecasts with minimal deviation from the observed test data, although it lacked the ability to capture trends in the series.

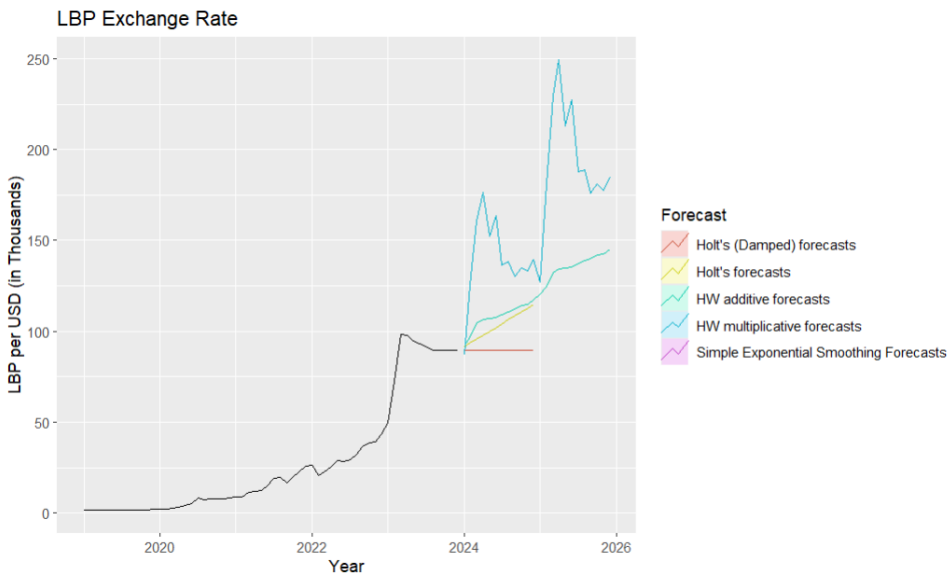


Figure 8: ETS Models Forecasts

ARIMAX Modeling for Exchange Rate Forecasting

ARIMAX (AutoRegressive Integrated Moving Average with Explanatory Variables) is an extension of the ARIMA model that incorporates external regressors to improve forecasting accuracy. The ARIMAX model combines the core components of ARIMA (AR, I, and MA) with linear regression terms that account for the influence of external variables.

The general ARIMAX(*p,d,q*) model can be expressed as:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} + \beta_1 X_{1t} + \beta_2 X_{2t} + \cdots + \beta_k X_{kt}$$

In this study, the ARIMAX model was applied to forecast the Lebanese Pound (LBP) exchange rate against the US Dollar using foreign currency reserves (FCB) and the Consumer Price Index (CPI) as regressors.

Model Estimation

The ARIMAX model was fitted using the Box-Cox transformed training data. The external regressors, FCB and CPI, were also Box-Cox transformed to stabilize variance. The best-fitting model was determined automatically using auto.arima() with the transformed exchange rate as the dependent variable and the transformed FCB and CPI as regressors.

Parameter	Estimate	Standard Error	95% Confidence Interval
AR (1)	0.7090	0.0991	[0.515, 0.903]
MA (1)	0.4002	0.1386	[0.126, 0.674]
Intercept	26.2308	12.1816	[2.336, 50.125]
Foreign currency Reserves	-21.3861	12.6689	[-46.228, 3.456]
CPI	1.8174	0.2990	[1.221, 2.414]

Table 4: ARIMAX Coefficients

The ARIMAX model selected had ARIMA(1,1,1) errors, with a log-likelihood of 7.79, AIC of -3.58, and BIC of 8.99.

The ARIMAX model incorporates key parameters to explain the exchange rate's dynamics with external regressors. The **AR(1)** coefficient (0.7090.7090.709) indicates a strong dependence on the previous value, while the **MA(1)** term (0.40020.40020.4002) reflects the influence of past forecast errors.

The **intercept** (26.230826.230826.2308) represents the baseline of the transformed series. Among the regressors, **foreign currency reserves** have a negative coefficient (−21.3861-21.3861−21.3861), suggesting that higher reserves reduce exchange rate volatility. Conversely, the **Consumer Price Index (CPI)** has a positive coefficient (1.81741.81741.8174), indicating that inflation increases the exchange rate. Most coefficients are statistically significant based on their 95% confidence intervals, except for reserves, which marginally overlaps zero.

Forecasting and Evaluation

The ARIMAX model was used to forecast the LBP/USD exchange rate for 12 months, using FCB and CPI as external variables. The forecasts were back-transformed to the original scale using the inverse Box-Cox transformation.

Metric	Value
Mean Absolute Error (MAE)	4.8331
Root Mean Squared Error (RMSE)	5.2577

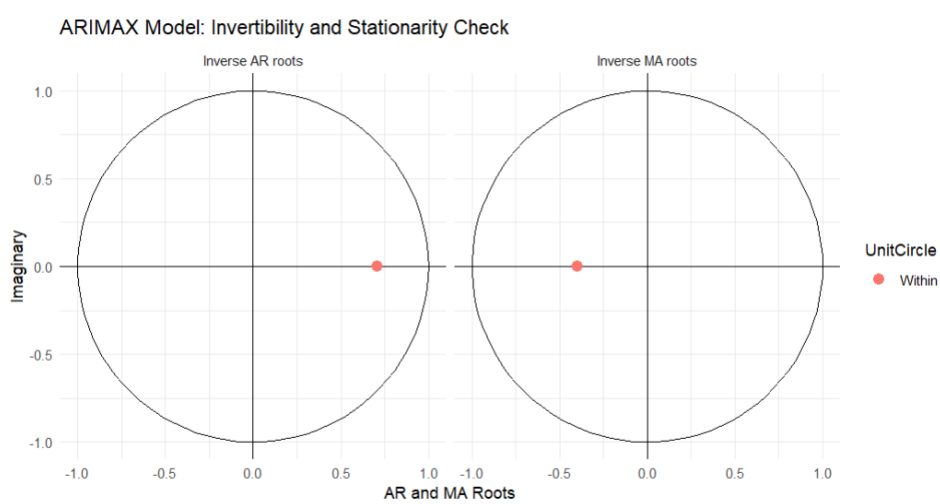
Table 5: Accuracy Measures for ARIMAX

Comparison with ARIMA and ETS Models

The ARIMAX model outperformed both ARIMA and ETS models in forecasting accuracy. While ARIMA struggled to incorporate external factors and ETS showed limitations during volatile periods, ARIMAX leveraged meaningful regressors like FCB and CPI to capture additional dynamics of the exchange rate.

Model	MAE	RMSE	AICc
ARIMA	24.174	28.109	18.064
ETS (Holt's)	13.513	15.300	441.930
ARIMAX	**4.833**	**5.257**	**-3.580**

Table 6: Comparison of ARIMA and ETS



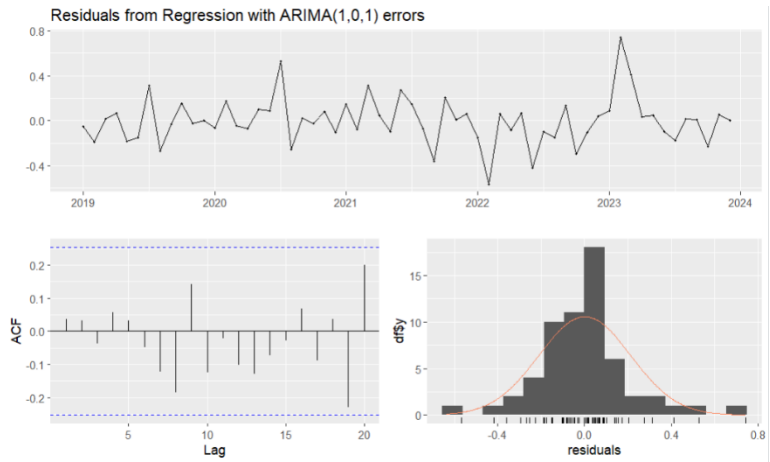


Figure 9: ARIMAX Residuals

The residuals of the ARIMAX model were analyzed using the Ljung-Box test to assess whether they are white noise. The test returned a *p*-value of 0.6631 (**Q*=7.649, d*f*= 10**), which is greater than the significance level of 0.05. This indicates that the residuals are uncorrelated, and the model has adequately captured the underlying structure of the data. Additionally, the invertibility check confirmed that all AR and MA roots lie within the unit circle, verifying the model's stationarity and invertibility.

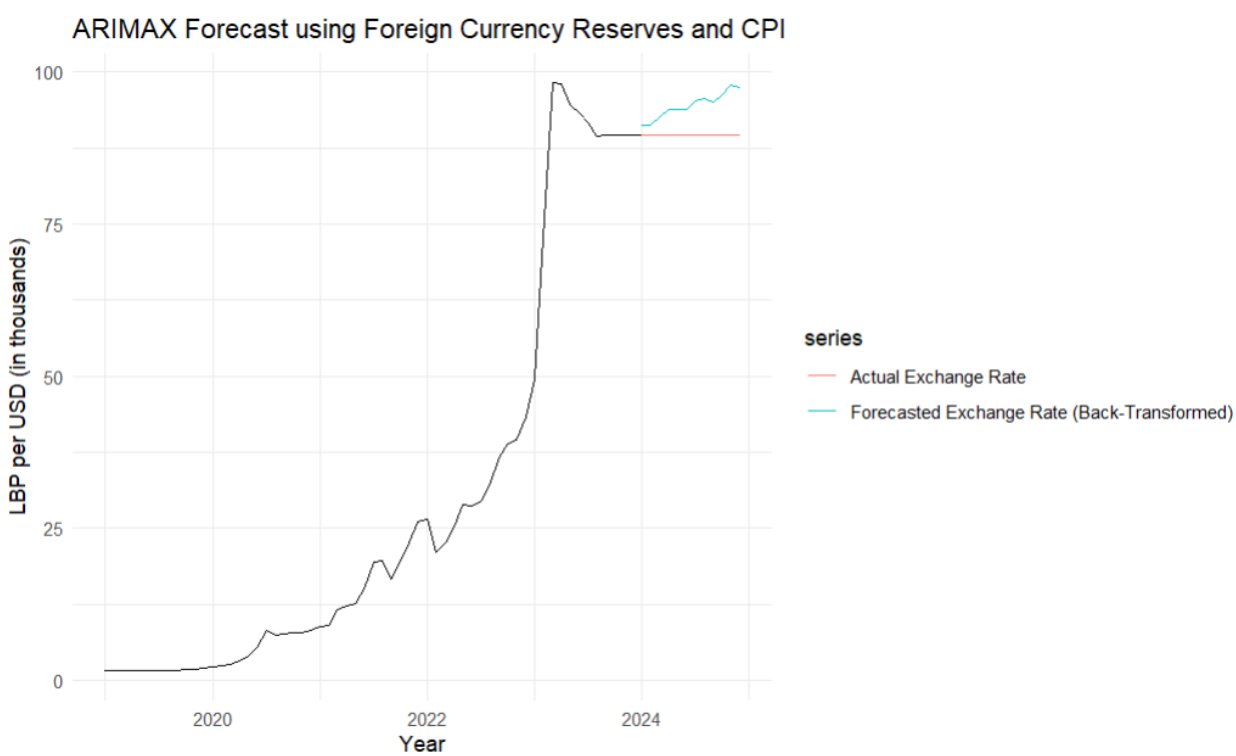


Figure 10: ARIMAX Forecast

Conclusion

The ARIMAX model, incorporating foreign currency reserves and CPI as regressors, demonstrated superior performance in forecasting the LBP/USD exchange rate compared to ARIMA and ETS models. The inclusion of external variables significantly improved accuracy, highlighting the importance of multivariate approaches in capturing complex economic relationships. This model provides valuable insights for policymakers and analysts managing currency fluctuations.

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